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ORIGINAL RESEARCH ARTICLE

Relating American Football Age of First Exposure to Patient-Reported Outcomes and Medical Diagnoses Among Former National Football League Players: An NFL-LONG study

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Abstract

Background The age of first exposure (AFE) to American football participation is a growing concern for late-life function. Mixed evidence exists surrounding AFE and may be attributed to varied methods employed across studies.

Objective To examine the associations between AFE to American football participation with measures of cognitive, behavioral, and physical function and brain-related medical diagnoses across age categories among former National Football League players.

Methods We conducted a cross-sectional survey of 1784 former players (age: 52.3 ± 16.3 years, AFE: 11.3 ± 2.9 years, years of football: 17.5 ± 4.5 years, $86.9\% \geq$ one lifetime concussion). Players completed a general health questionnaire recording demographics, football playing history (including AFE), and diagnoses (anxiety, depression, any form of dementia, mild cognitive impairment). Players completed Patient-Reported Outcomes Measurement Information System (PROMIS) measures assessing domains of cognitive and physical function, anxiety, depression, sleep disturbance, pain interference, and emotional-behavioral dyscontrol. Multivariable linear and binomial regression models were used to examine the associations of AFE and age with PROMIS outcomes and diagnoses, respectively.

Results No significant AFE by age interactions were detected for PROMIS outcomes ($p \geq 0.066$) or diagnoses ($p \geq 0.147$). Younger AFE associated with higher PROMIS scores of anxiety ($B = -0.22$, $p = 0.016$), depression ($B = -0.22$, $p = 0.010$), sleep disturbances ($B = -0.16$, $p = 0.007$), pain interference ($B = -0.19$, $p = 0.014$), and emotional-behavioral dyscontrol ($B = -0.22$, $p = 0.019$). Age was associated with all PROMIS outcomes ($p \leq 0.042$). AFE was not associated with the prevalence of anxiety, depression, dementia, or mild cognitive impairment ($p \geq 0.449$), while age was ($p \leq 0.013$).

Conclusions AFE was significantly associated with PROMIS outcomes, albeit low-strength associations (i.e., effect sizes), but not with diagnoses. Our findings indicate AFE is a significant but minor contributing factor for health-related quality of life in this cohort. Future work should incorporate additional characterizations of cumulative head impacts and related factors when examining long-term outcomes associated with football participation.

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Key Points

Age of first exposure (AFE) did not demonstrate any significant interaction effects with age when examining late-life function and medical diagnosis prevalence, which suggests any AFE effect does not appear have an aging-dose response.

AFE was weakly associated with self-reported function and was not associated with medical diagnoses among former NFL players, and age had greater independent associations with self-reported function and diagnoses than AFE.

Stakeholders (e.g., clinicians, individuals who have played football, family members) may not need to be overly concerned with the age they started playing football and their viewed cognitive and physical function.

1 Introduction

Tackle football is one of the most common sports among male youth athletes in the United States [1]. However, repetitive head impacts experienced by athletes playing football have led to growing concerns regarding youth participation [2–5] and particularly the age at which youth are first exposed to full contact play (age of first exposure (AFE)). AFE is theorized to be a proxy measure for repetitive head impacts experienced during a vulnerable period of brain maturation [6]. Earlier AFE concerns have been driven by prior work involving former National Football League (NFL) players reporting associations between younger AFE and late-life psychiatric and cognitive difficulties [6–9].

The examination of AFE has grown beyond just those who participate in professional football and now includes studies of former high school football players [7, 10–13], collegiate football players [7, 10, 11, 13–15], and contact sport participants in general [16–18]. Those examining AFE among various discrete age groups have sought to better understand the temporal risk for possible negative health consequences from football participation. Most studies of college-aged former athletes report no associations between AFE and psychological, neurologic, cognitive, behavioral, or physical function outcomes [10, 14, 15, 17, 18]. However, less is known about health-related outcomes among middle-aged former football players [11, 12, 19]. Specifically, minimal understanding exists surrounding diagnosis prevalence related to AFE and their relationship to individuals in their mid- to late-life stages. If earlier AFE can be equated to greater accumulation of repetitive head impacts during a neurodevelopmentally vulnerable period and influence neurobehavioral function [6–9], it is plausible earlier AFE may also contribute to a greater risk of neurodegenerative disease prevalence.

To date, AFE has not been examined in relation to the prevalence of specific medical diagnoses beyond chronic traumatic encephalopathy (CTE) [8]. Outside of inquiries related to AFE, the literature examining football participation and neurodegenerative disease is also mixed. Some studies have indicated those participating in football have greater rates of neurodegenerative diseases (Alzheimer's, amyotrophic lateral sclerosis) relative to the general population [20], while others found no difference [21, 22] or a lower risk [23, 24] of mortality and presence of known neurodegenerative disease risk factors. The mixed findings [10–12, 14–19, 25] leaves individuals uncertain about football playing safety.

Varying populations and research methods add complexity to the challenge of understanding how AFE is a risk factor for neurobehavioral difficulties [26]. Specifically, operationalization of the AFE metric (e.g., median-split vs. continuous), heterogeneity in function across cohorts recruited for study participation, inconsistent recruitment methods across studies, and the age groups examined in each study may contribute to the disparate findings [26]. These study differences may partially contribute to the mixed associations among older cohorts who enrolled participants with different age ranges and self-reported physical and cognitive impairments upon recruitment [6–8, 10–13, 19, 25]. Age is likely a modifying effect for understanding AFE, as age is one of the largest risk factors for neurodegenerative pathology [27, 28] and change in cognitive function [29]. Thus, it is important to account for the effect of age when trying to determine the independent association between AFE and outcomes.

The purpose of our study was to examine the associations between AFE and measures of self-reported function and self-reported diagnoses, while accounting for the effects of age. We hypothesized a significant AFE-by-age interaction would be observed for self-reported function outcomes and diagnoses such that AFE associations would not be observed for younger athletes while being observed for older athletes. We also hypothesized age would be associated with both function and diagnoses.

2 Methods

We conducted a cross-sectional study of former NFL players using self-administered questionnaires online or via paper hardcopy. This study was conducted as part of a larger, multiphasic longitudinal project called "Neurologic Function

across the Lifespan: A Prospective, LONGitudinal, and Translational Study for Former National Football League Players" (NFL-LONG) [30–33].

2.1 Participants

Former NFL players were recruited between January 2019 and February 2020 via email or mailed letters sent through listservs from the NFL Alumni Association or to individual teams or players through existing connections. Players were included if they participated in at least one NFL season. A total of 2,030 hard copy or electronic surveys were collected, with 236 excluded for being duplicate records from players and an additional ten players excluded for not providing identifying information [33]. A total of 1784 former NFL players were initially included in the study as previously described [33]. Language describing the investigatory nature of the questionnaire was provided at the outset. Completion of the questionnaire after the introductory paragraph implied consent, with the study protocol approved by our Institutional Review Board (IRB).

2.2 Questionnaires

The self-administered general health questionnaire was adapted and modified from a previously established questionnaire [34, 35] designed to comprehensively assess demographics (e.g., age, race, education), detailed football playing and concussion history, medical history and diagnoses, musculoskeletal injury history, current health status, current nutrition, substance use, current health-related habits, and self-reported cognitive, behavioral, and physical function [33]. Five former NFL players also piloted, reviewed, and provided feedback to ensure clear question-and-response formatting.

2.2.1 Age of First Exposure (AFE)

AFE was derived from the survey item asking participants "At what age (in years) did you start playing organized football?" Previous research has characterized AFE as: < 12 and ≥ 12 years [6–8, 10–12, 14, 16, 17, 25], based upon an individual's school level at the time of AFE (e.g., before high school vs. during or after high school) [13, 19], and as a continuous or discrete variable [7, 8, 11, 12, 15, 16, 18, 25, 36]. We examined AFE as a discrete variable as the NFL-LONG dataset yielded a Gaussian distribution (see Sect. 3, Results). Further, discrete characterization provided greater statistical power and temporal resolution to detect AFE's effect on outcomes, compared to other potential characterizations [37, 38].

2.2.2 Age

Age was self-reported and categorized into five different decades of life (20–29 years old, 30–39 years old, 40–49 years old, 50–59 years old, 60–69 years old) and a 70–99-year-old group. Participants aged 70–99 years were grouped together to avoid using small group sizes that would reduce statistical power. Aging effects are often not linear across the lifespan for function or disease onset [39–41]. Therefore, age at the time of completing the general health questionnaire was operationalized in this manner to statistically account for non-linear effects by serving as a categorical grouping variable rather than a quantitative variable.

2.2.3 Self-Reported Diagnoses

Participants were asked via binary (yes or no) items whether a physician or other healthcare provider had told them they have or had any of the following separate diagnoses: anxiety, depression, dementia-related disease (Alzheimer's disease, Parkinson's disease, frontotemporal dementia, Lewy-body dementia, vascular dementia, other dementia), or mild cognitive impairment. These diagnoses were selected because they may alter neurological and cognitive-behavioral function [6–9].

2.2.4 Self-Reported Function Outcomes

The Patient-Reported Outcomes Measurement Information System (PROMIS) measures were developed with National Institutes of Health (NIH) support to ensure reliable and valid construct measurements, and represented some of the most validated measures to date [42–45]. For this study, we used the PROMIS Adult Cognitive Function short form-4a [42], four, four-item domains of the PROMIS-29 Profile V2.0 (depressive symptoms, anxiety, physical function, and sleep disturbances) [43], and the Quality of Life in Neurological Disorders (Neuro-QOL) Emotional and Behavioral Dyscontrol form [45]. The NFL-LONG study prospectively used these measures to comprehensively assess self-reported cognitive, behavioral, and physical function among former NFL players due to prior work indicating potential associations between AFE and these PROMIS constructs [6–9].

All PROMIS and Neuro-QOL outcomes were derived from individual items that were scored on a 5-point Likert scale, summed, and then transformed to T-scores per standard use [42–45]. Each PROMIS and Neuro-QOL outcome T-score is interpreted with a score of 50, representing the US population normative mean and a standard deviation of 10. Higher scores indicate stronger construct presence (i.e., greater cognitive function, anxiety, depression, emotional-behavioral dyscontrol, physical function, sleep disturbances). Scores between 45 and 80 are considered within

normal limits for cognitive function and physical function, and scores between 20 and 55 are considered within normal limits for depression, anxiety, emotional-behavioral dyscontrol, and sleep disturbances [46].

2.3 Statistical Analysis

Descriptive statistics were calculated for age, years of football participation, race, education, concussion history, AFE, and the prevalence of the following diagnoses: anxiety, depression, any form of dementia-related disease, and mild cognitive impairment.

Multivariable linear regression models were used to estimate the association between AFE and each self-reported function outcome while accounting for age as a potential confounder alongside their interaction term. Unstandardized beta-coefficients (B), 95% confidence intervals (95% CI), and coefficients of variation (R^2) were calculated for the multivariable linear regression models with listwise deletion occurring in each individual model in the event of case data missingness. All unstandardized B with 95% CIs excluding 0.00 were considered statistically significant. The AFE and age interaction term was examined from the omnibus Type III tests via F tests, and if significant interactions were not present ($p \geq 0.05$), the interaction term was dropped.

Multivariable Poisson regressions were used to calculate the prevalence ratios (PRs) of diagnoses based on AFE while accounting for age and their interaction. Listwise deletion occurred in each individual model in the event of case data missingness. All Poisson regression models used a robust sandwich covariance estimator for model fitting as this approach yields conservative estimates over binomial modeling when convergence criteria are not met [47, 48]. All PRs calculated with associated 95% CIs excluding 1.00 were considered statistically significant. The AFE and age interaction term was examined from the omnibus type III tests via likelihood ratio tests, with non-significant interactions ($p \geq 0.05$) being dropped from the model.

The following covariates were assessed for model inclusion based upon prior work with this cohort [33]: body mass index, learning disability, race, marital status, drug or alcohol use, exercise frequency, sleep duration, depression diagnosis, anxiety diagnosis, and education. These covariates were only used if our a priori Pearson r or Spearman ρ coefficients threshold of 0.20 was met or exceeded between the covariate and AFE. Effect size estimates rather than p value thresholds were used due to the large sample size and thus likelihood for including statistically significant covariates with low clinical meaning into the model [49]. The relationship between AFE and concussion history as an ordinal variable ($\rho = 0.05$) and total years of football participation as a continuous variable ($r = -0.57$) were initially considered, but raised endogeneity concerns as earlier concussion history and total years of football participation are partially dependent upon AFE [50]. Covarying for concussion history and total years of football participation, therefore, is likely to reduce the true effect estimate of AFE (i.e., show a weaker effect of AFE than is true) and improperly address our specific aims. Therefore, we did not adjust for concussion history or total years of football participation.

All model assumptions (residual normality, linearity, homogeneous variance, multicollinearity, influential observations) were visually assessed via graphical plots and confirmed prior to reporting, and statistical analyses were performed using the R Foundation for Statistical Programming (version 4.0.4; Vienna, Austria).

3 Results

The sample of 1784 former NFL players had a mean age of 52.3 ± 16.3 years and the participant frequency across each decade of life (i.e., age) was: 20–29 years old ($n = 150$), 30–39 years old ($n = 360$), 40–49 years old ($n = 285$), 50–59 years old ($n = 321$), 60–69 years old ($n = 360$), 70–99 years old ($n = 308$). Participants played a mean of 17.5 ± 4.5 years of football, of which 5.8 ± 4.0 years (median = 5 years) were played at the professional level. Approximately 58.4% were White/non-Hispanic, 84.8% had at least a Bachelor's degree, and 86.9% had a history of concussion (see Walton et al. [33] for more demographic details). AFE demonstrated a Gaussian distribution (skewness = − 0.19, kurtosis = − 0.72) with a mean AFE of 11.3 ± 2.9 years (median = 12 years; range 4–18 years). Descriptive statistics for all self-reported function outcomes and diagnosis outcomes by age and overall are presented in Table 1, with each PROMIS and Neuro-QOL measure mean score being within 1 standard deviation of the USA general population in each decade of life.

3.1 AFE and Self-reported Function Outcomes Across Age

Figure 1 illustrates the PROMIS and Neuro-QOL outcomes for each AFE year across age, and Fig. 2 illustrates the PROMIS and Neuro-QOL outcomes across age. No significant omnibus AFE by age interactions were observed for any PROMIS or Neuro-QOL outcomes ($p \geq 0.066$) and therefore all interaction terms were dropped from models. All multivariable models were overall statistically significant, explaining a small amount of overall variance ($p < 0.001$; R^2 range 0.03–0.06; Table 2) with varying univariable effects.

Main effects for AFE were observed such that every 1-year delay in AFE was associated with decreased (i.e., less presence) PROMIS and Neuro-QOL scores for anxiety: $B = -0.22$, $p = 0.016$; depression: $B = -0.22$, $p = 0.010$;

Table 1 Summary statistics for self-reported function and diagnoses by age

Self-reported function outcomes	All groups combined	20–29 years old	30–39 years old	40–49 years old	50–59 years old	60–69 years old	70–99 years old
	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)
PROMIS-cognitive ($n=1749$)	45.34 (10.22)	49.56 (9.78)	46.70 (10.04)	43.73 (10.00)	44.35 (10.57)	44.86 (10.15)	44.76 (9.90)
PROMIS-anxiety ($n=1728$)	53.59 (10.58)	53.92 (10.56)	55.72 (10.41)	56.84 (10.50)	54.43 (10.81)	51.09 (9.90)	49.70 (9.59)
PROMIS-depression ($n=1737$)	50.81 (9.97)	49.96 (10.31)	51.94 (10.27)	53.85 (10.20)	51.79 (10.23)	48.74 (8.86)	48.39 (9.09)
PROMIS-sleep disturbance ($n=1686$)	52.55 (6.48)	51.75 (5.80)	52.81 (6.45)	53.91 (6.95)	53.47 (7.10)	52.13 (6.45)	50.69 (4.98)
PROMIS-physical function ($n=1702$)	45.54 (8.91)	50.92 (7.43)	47.64 (8.57)	44.87 (8.77)	43.74 (8.72)	45.15 (8.65)	43.15 (9.04)
PROMIS-pain interference ($n=1753$)	56.34 (9.26)	51.17 (8.39)	54.18 (9.20)	58.11 (9.37)	58.67 (9.31)	56.99 (8.86)	56.59 (8.60)
Neuro QOL-EBD ($n=1750$)	51.11 (10.84)	49.62 (11.38)	52.30 (11.10)	54.22 (11.41)	52.08 (10.75)	49.31 (10.00)	48.66 (9.75)
Diagnosis[a]	*n* (%)	*n* (%)	*n* (%)	*n* (%)	*n* (%)	*n* (%)	*n* (%)
Anxiety	333 (18.67)	16 (10.67)	80 (22.22)	62 (21.75)	72 (22.43)	61 (16.94)	42 (13.63)
Depression	357 (20.01)	16 (10.67)	70 (19.44)	73 (25.61)	75 (23.36)	67 (18.61)	56 (18.18)
Any dementia	107 (6.00)	0 (0.00)	6 (1.67)	8 (2.81)	19 (5.92)	35 (9.72)	39 (12.66)
Mild cognitive impairment	338 (18.95)	5 (3.33)	44 (12.22)	48 (16.84)	75 (23.36)	85 (23.61)	80 (25.97)

[a]Diagnosis values presented as sample count and proportion who reported "yes" to each specific diagnosis presence

sleep disturbances: $B=-0.16$, $p=0.007$; pain interference: $B=-0.19$, $p=0.014$; and emotional-behavioral dyscontrol: $B=-0.22$, $p=0.019$ (Table 2). No significant main effect for AFE was observed for the PROMIS cognitive or physical function measures ($p\geq 0.168$).

Omnibus main effects for the age groups were observed for all self-reported function outcomes (p values <0.001, Table 2). All age groups relative to the 20–29-year-old group reported worse PROMIS cognitive function, physical function, and pain interference that did not consistently change across age. PROMIS anxiety scores were significantly higher among the 30–39- and 40–49-year-olds compared to the 20–29-year-old participants, whereas PROMIS anxiety scores were lower among the 60–69- and 70–99-year-old groups when compared to the 20–29-year-old participants (Table 2). PROMIS depression scores were higher among the 30–39- through 50–59-year-old groups relative to the 20–29-year-old group, but did not differ between the 60–69- and 70–99-year-old groups relative to the 20–29-year-old group. PROMIS sleep disturbance scores were higher in the 40–49- and 50–59-year-old groups relative to the 20–29-year-old group (Table 2). Emotional-behavioral dyscontrol scores were higher in the 30–39- through 50–59-year-old groups compared to the 20–29-year-old group.

3.2 AFE Self-reported Diagnoses Across Age

No significant omnibus AFE-by-age interactions were observed for diagnoses (all p values ≥ 0.147; Table 1) and therefore dropped from all models. No AFE main effects were observed for diagnoses of anxiety, depression, dementia, or mild cognitive impairment ($p\geq 0.449$; Table 3). Omnibus main effects for age were present across all diagnoses ($p\leq 0.013$). A clinical diagnosis of anxiety was more prevalent among the 30–39- through 50–59-year-old groups than the 20–29-year-old group (Table 3). A clinical diagnosis of depression was more prevalent across all age groups compared to the 20–29-year-olds, but did not demonstrate a linear increase. Any form of dementia and mild cognitive impairment were more prevalent among the 50–59- through 70–99-year-old groups compared to the 30–39-year-old group (i.e., the reference group for dementia and mild cognitive impairment due to 20–29-year-olds having small diagnosis frequencies). No other differences in the prevalence of diagnoses were observed (Table 3).

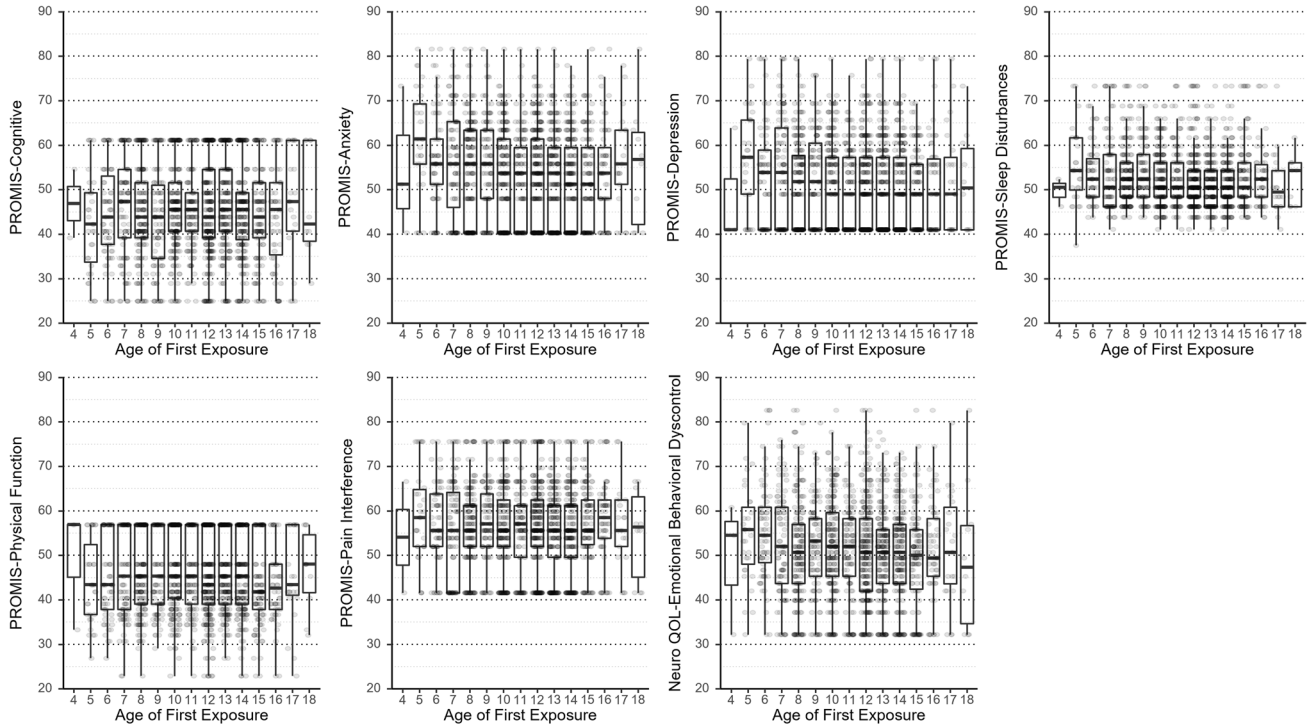

Fig. 1 Age of first exposure across self-reported function outcomes. Age of first exposure via standard Tukey boxplots and 10% weighted datapoints across each age level among the self-reported function outcomes. *PROMIS* Patient-Reported Outcomes Measurement Information System, *QOL* quality of life

4 Discussion

Prior work among college-aged football players and contact sport athletes has shown no associations between AFE and neurocognitive, psychological, or physical function [14, 17, 18]. However, work among mid- to late-life football participation among former professional and non-professional individuals has demonstrated mixed associations between earlier AFE and neurodegenerative outcomes [6–8, 10–13, 19, 25]. Our present study with a large sample of former NFL-players sheds light on the temporal relationships between AFE, global self-reported function, and diagnoses by examining these relationships across decades of life.

We observed earlier AFE had a significant association with greater anxiety, depression, sleep disturbances, pain interference, and emotional-behavioral dyscontrol. These statistical relationships between self-reported function outcomes and AFE were, however, weak as demonstrated by the observed B values (Table 2). Deducing the strongest independent effect of delaying AFE by 1 year ($B = -0.22$; Table 2) by the previously reported age threshold of 12 years [6–8, 10–12, 14, 16, 17, 25, 36] demonstrates a T-score difference of 2.6, and extending to 18 years (i.e., no football exposure throughout adolescence) indicates a *T*-score difference of 4.0. Interpreting this effect indicates the largest *T*-score difference associated with the earliest levels of play still fell within normal limits compared to the general population [46]. We also did not observe any main effects for AFE with diagnoses of anxiety, depression, any form of dementia, or mild cognitive impairment, suggesting AFE alone is not independently associated with these diagnoses.

Our work expands upon previous studies by examining whether an aging-dose response exists for AFE on self-reported function and diagnosis prevalence. Overall, we did not observe any interaction effects of AFE and age, indicating that any observed effect of AFE is not augmented across the lifespan. Our findings indicate AFE is a minor consideration regarding functional outcomes and brain-related diagnoses. The use of AFE, however, may be an inadequate proxy metric for understanding the association between risk for adverse long-term outcomes and exposure to repetitive head trauma. Regardless, modifiable factors such as exercise, diet, and sleep have been reported to have meaningful effects on self-reported function in former NFL players [33]. Thus, greater emphasis on current known modifiable factors may optimize player health while more granular repetitive head impact metrics are developed.

Normal aging is a well-established contributor to cognitive [39] and physical [40] decline as well as increased risk for neurologic dysfunction [41]. Therefore, the independent effect of age on PROMIS and Neuro-QOL outcomes (Table 2) and diagnoses of mild cognitive impairment and

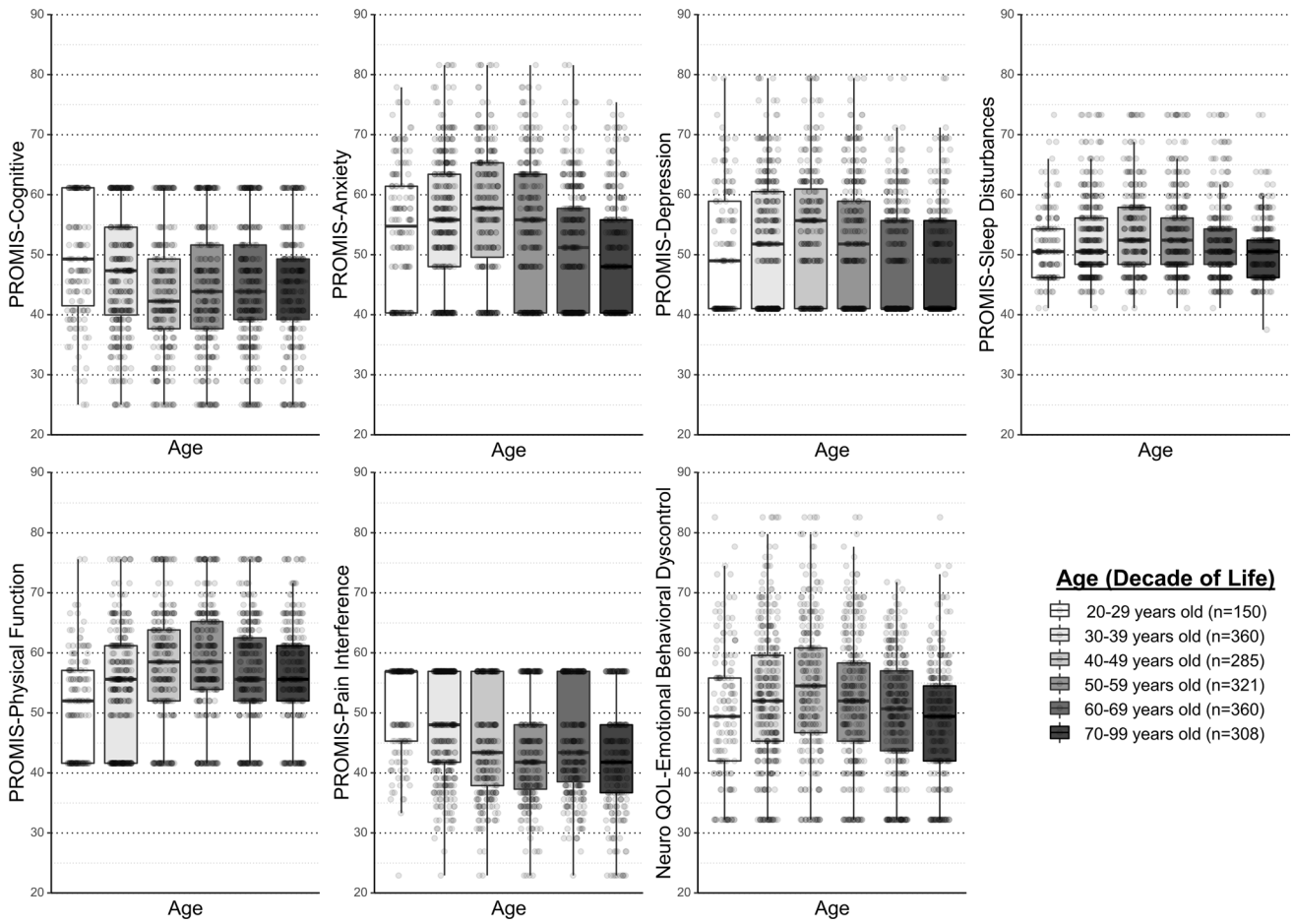

Fig. 2 Age (decade of life) across self-reported function outcomes. Age via standard Tukey boxplots and 10% weighted datapoints across each group among the self-reported function outcomes. *PROMIS* Patient-Reported Outcomes Measurement Information System, *QOL* quality of life

any form of dementia among the older groups relative to the 20–29-year-old group was anticipated (Table 3). The mean scores for all age groups fell within the normal limits of the general US population [42–45], though fluctuations in participant distributions were observed (Fig. 2). The 40–49- and 50–59-year-old groups reported the least adaptive self-reported function across all domains, except for physical function, compared to the 20–29-year-old groups (Table 2). It is unclear what may be contributing to the notably lower function among the 40–49- and 50–59-year-old groups. Our study does not allow for any causal inferences, but speculatively, the older 60–69- and 70–99-year-old cohorts may represent survivorship bias, where individuals in this age group are approaching, meeting, or exceeding typical life expectancy [51] and therefore may have started from a higher functional level than currently reported. Our findings highlight the importance of accounting for age when examining AFE, as well as self-reported function outcomes.

Our study had inherent and noteworthy limitations. First, the outcomes were self-reported by the participants and may be subject to recall bias. Future research might seek to determine how well the self-reported function measures relate to objective measures of the same constructs to provide a more comprehensive understanding of these relationships. Furthermore, participation was voluntary and, therefore, the findings may be affected by ascertainment bias. The self-report survey nature and biases at play may have impacted data pertaining to measures, such as medical history that were self-reported by participants [52]. It will be useful for future research to validate self-reported diagnosis accuracy or obtain confirmed medical provider documentation, although both strategies will likely fall beyond survey-based research. Our analyses also used standard group-level statistical modeling that may give biased perception to late-life function, and thus we provided box and whisker plots with individual datapoints

Table 2 Relationship between age of first exposure (AFE) and self-reported function outcomes by age

	Overall model	AFE 1-year increase		Age (decade of life) 20–29 years old		30–39 years old		40–49 years old		50–59 years old		60–69 years old		70–99 years old	
	Intercept	*B*	*p* value	*B*	*p* value	*B*	*p* value	*B*	*p* value	*B*	*p* value	*B*	*p* value	*B*	*p* value
	R^2	(95% CI)		(95% CI)		(95% CI)		(95% CI)		(95% CI)		(95% CI)		(95% CI)	
PROMIS-cognitive ($n = 1699$)[a]	48.24	0.14	0.116	Ref	–	− 2.93	0.003[b]	− 6.05	< 0.001[b]	− 5.41	< 0.001[b]	− 5.00	< 0.001[b]	− 5.22	< 0.001[b]
	0.03	(− 0.03, 0.31)				(− 4.88, − 0.97)		(− 8.08, − 4.02)		(− 7.41, − 3.41)		(− 6.98, − 3.01)		(− 7.30, − 3.14)	
PROMIS-anxiety ($n = 1683$)[a]	55.94	− 0.22	0.016[b]	Ref	–	2.01	0.046[b]	3.32	0.002[b]	0.86	0.405	− 2.38	0.020[b]	− 3.37	0.002[b]
	0.06	(− 0.40, − 0.04)				(0.04, 3.99)		(1.27, 5.38)		(− 1.17, 2.89)		(− 4.39, − 0.38)		(− 5.49, − 1.26)	
PROMIS-depression ($n = 1689$)[a]	52.21	− 0.22	0.010[b]	Ref	–	2.05	0.033[b]	4.10	< 0.001[b]	1.95	0.049[b]	− 0.83	0.395	− 0.79	0.440
	0.04	(− 0.39, − 0.05)				(0.16, 3.94)		(2.14, 6.06)		(0.01, 3.88)		(− 2.74, 1.08)		(− 2.81, 1.22)	
PROMIS-sleep disturbance ($n = 1637$)[a]	53.35	− 0.16	0.007[b]	Ref	–	1.09	0.084	2.28	< 0.001[b]	1.87	0.004[b]	0.65	0.315	− 0.68	0.320
	0.03	(− 0.27, − 0.04)				(− 0.15, 2.34)		(0.98, 3.57)		(0.59, 3.15)		(− 0.62, 1.92)		(− 2.03, 0.66)	
PROMIS-physical Function ($n = 1650$)[a]	50.7	0.01	0.860	Ref	–	− 3.20	< 0.001[b]	− 6.00	< 0.001[b]	− 7.04	< 0.001[b]	− 5.64	< 0.001[b]	− 7.68	< 0.001[b]
	0.06	(− 0.14, 0.16)				(− 4.87, − 1.52)		(− 7.74, − 4.27)		(− 8.75, − 5.32)		(− 7.35, − 3.94)		(− 9.47, − 5.89)	
PROMIS-pain interference ($n = 1703$)[a]	53.13	− 0.19	0.014[b]	Ref	–	3.11	< 0.001[b]	7.13	< 0.001[b]	7.62	< 0.001[b]	6.01	< 0.001[b]	5.85	< 0.001[b]
	0.06	(− 0.34, − 0.04)				(1.38, 4.85)		(5.33, 8.93)		(5.85, 9.40)		(4.25, 7.77)		(4.00, 7.69)	
Neuro QOL-EBD ($n = 1704$)[a]	51.63	− 0.22	0.019[b]	Ref	–	2.93	0.005[b]	5.04	< 0.001[b]	2.80	0.009[b]	0.11	0.914	− 0.27	0.806
	0.04	(− 0.40, − 0.04)				(0.88, 4.97)		(2.90, 7.17)		(0.70, 4.90)		(− 1.97, 2.20)		(− 2.44, 1.90)	

All multivariable models were statistically significant ($p < 0.001$) and initially ran with an AFE-by-age interaction, but no significant interactions were observed and therefore interaction terms were dropped from all models

AFE age of first exposure to football, *B* unstandardized beta coefficient, *PROMIS* Patient-Reported Outcomes Measurement Information System, *QOL* quality of life, *EBD* emotional-behavioral dyscontrol, *Ref* reference group

[a]There was some missingness in PROMIS outcomes resulting in the sum of all age groups not equaling the total self-reported function model sample size

[b]Indicates statistical significance ($p < 0.05$)

Table 3 Relationship between age of first exposure and diagnoses by age

	AFE		Age (decade of life)											
	1-year increase		20–29 years old		30–39 years old		40–49 years old		50–59 years old		60–69 years old		70–99 years old	
	PR (95% CI)	*p* value	PR (95% CI)	*p* value	PR (95% CI)	*p* value	PR (95% CI)	*p* value	PR (95% CI)	*p* value	PR (95% CI)	*p* value	PR (95% CI)	*p* value
Anxiety ($n = 1731$)[a]	0.99 (0.95, 1.02)	0.449	Ref	–	2.07 (1.25, 3.41)	0.005[c]	2.02 (1.21, 3.37)	0.007[c]	2.11 (1.27, 3.50)	0.004[c]	1.64 (0.98, 2.76)	0.060	1.35 (0.78, 2.34)	0.285
Depression ($n = 1731$)[a]	1.00 (0.97, 1.04)	0.978	Ref	–	1.78 (1.06, 2.96)	0.027[c]	2.39 (1.44, 3.94)	0.001[c]	2.21 (1.34, 3.65)	0.002[c]	1.74 (1.04, 2.91)	0.034[c]	1.75 (1.03, 2.95)	0.037[c]
Any dementia ($n = 1731$)[a]	1.02 (0.94, 1.10)	0.590	–[b]	–[b]	Ref	–	1.97 (0.65, 5.98)	0.231	4.05 (1.51, 10.91)	0.006[c]	6.56 (2.54, 16.93)	< 0.001[c]	8.75 (3.33, 23.01)	< 0.001[c]
Mild cognitive impairment ($n = 1731$)[a]	0.99 (0.95, 1.03)	0.526	–[b]	–[b]	Ref	–	1.44 (0.98, 2.12)	0.061	2.00 (1.41, 2.83)	< 0.001[c]	2.01 (1.43, 2.84)	< 0.001[c]	2.30 (1.62, 3.26)	< 0.001[c]

All multivariable models were initially run with an AFE-by-age interaction, but no significant interactions were observed and therefore interaction terms were dropped from all models

PR prevalence ratio, *AFE* age of first exposure to football

[a]Some missingness in diagnosis reporting resulted in the sum of yes/no diagnosis not equaling the total sample size of 1784

[b]20-29-year-old cohort excluded from model due to insufficient cases with associated diagnoses ($n \leq 5$)

[c]Indicates statistical significance ($p < 0.05$)

(Fig. 2) to illustrate some of the individuals contributing to the groups did report function below normal limits in line with the typical population distributions. Further, causal inferences cannot be drawn from this study and the findings reported should not be treated as such. Lastly, the aging comparisons across decades of life should be considered cross-sectional snapshots in time and not as longitudinal examination of individuals, with this approach to categorizing age being a delimitation.

5 Conclusions

AFE did not demonstrate any significant interaction effects with age when examining late-life function and medical diagnosis prevalence, which suggests any AFE effect does not appear to have an aging-dose response. Earlier AFE (i.e., starting football earlier in life) demonstrated statistically significant negative associations with self-reported cognitive impairment, anxiety, depression, sleep disturbances, physical function, pain interference, and emotional-behavioral dyscontrol. Though statistically significant, these AFE effects were small in magnitude and suggest AFE is not a primary driver of long-term functioning when compared to other known variables [33]. AFE was not independently associated with late-life medical diagnoses or self-reported function. Age demonstrated substantially greater independent associations with all self-reported functional outcomes, which is generally consistent with typical effects of aging on late-life function. Future work may explore whether AFE is an appropriate proxy measure for adolescent cumulative head impacts to better determine long-term football safety.

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Declarations

Author contributions All authors agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. Specific author contributions are as follows: LBL: established research hypotheses, planned analyses of the data, interpretation of study findings, manuscript preparation and revision, final approval of the submitted manuscript. SRW, BLB, AC, and ZYK: creation and distribution of the study survey, management of study database, established research hypotheses, planned analyses of the data, interpretation of study findings, manuscript preparation and revision, final approval of the submitted manuscript. JDD, RCM, RJE, MAM, KMG, and WPM III: creation and distribution of the study survey, interpretation of study findings, manuscript preparation and revision, final approval of the submitted manuscript. Other contributors to the project are as follows (included in acknowledgements): Candice Goerger, Project Administrator: Assisted in the creation and distribution of the survey instrument as well as collection and management of the survey responses. Additionally, she was responsible for overseeing the project administration and execution. Hope Campbell, Project Coordinator (Medical College of Wisconsin): Assisted in the creation and distribution of the survey instrument as well as collection and management of the survey responses. Additionally, she assisted with oversight of the project administration and execution. Caprice Hunt, Project Coordinator (University of North Carolina at Chapel Hill): Assisted in the creation and distribution of the survey, oversight of the project administration, and execution of the study.

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Availability of data and material The dataset is not available for public sharing or upon individual's reasonable requests due to privacy and sharing restrictions from the IRB.

Research involving human participants The study was performed in accordance with the standards of ethics outlined in the Declaration of Helsinki.

Informed consent All study procedures were reviewed and approved by the University of North Carolina-Chapel Hill's IRB, as well the local IRB at each of the performance sites. Completion of the questionnaire after the introductory paragraph implied consent.

Code availability Not applicable.

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